



Liver & Extrahepatic Biliary Apparatus

Histology GIT Block · MEDucation by Mattin Majid

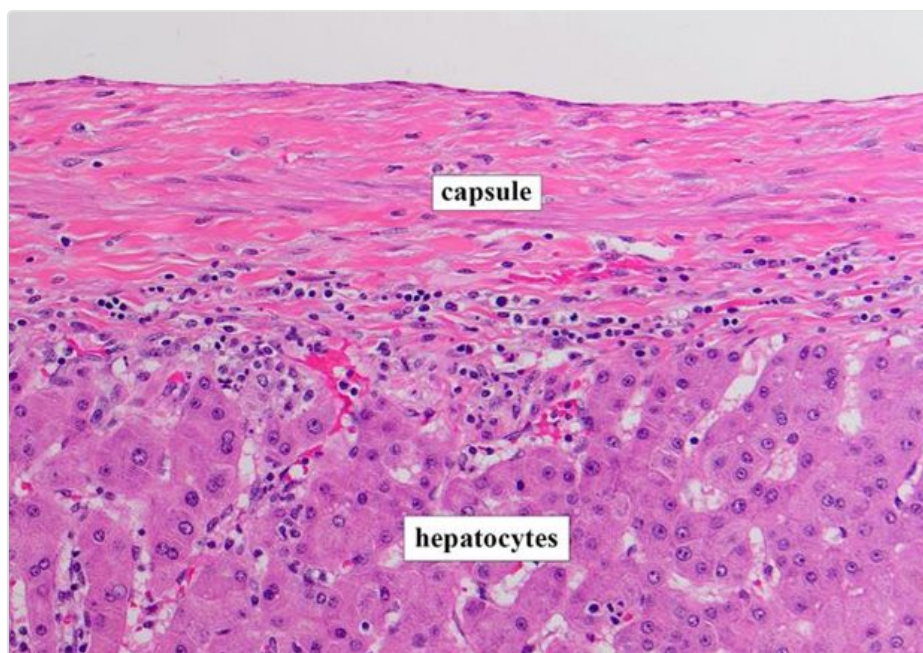
BEFORE YOU READ — ANSWER FROM MEMORY

1. What structure sits at each of the six corners of a classic hepatic lobule, and what makes it up?
2. Which liver cell lives in the space of Disse and stores vitamin A?
3. Trace bile from the bile canaliculus all the way to the duodenum.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Liver — overview

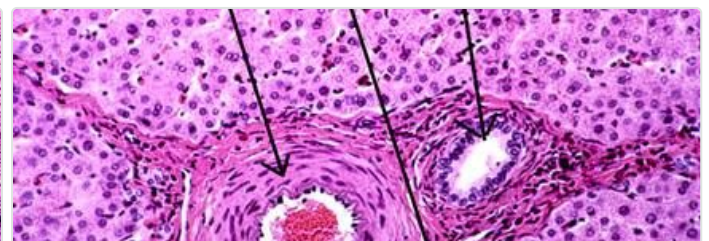
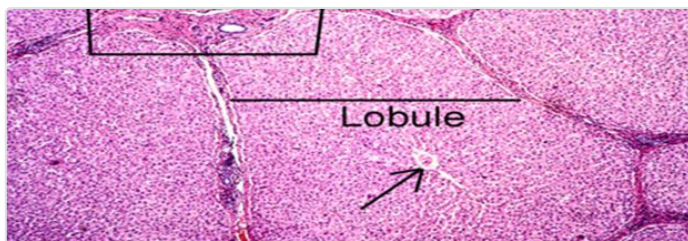
Location	Largest gland in the body; abdominal cavity beneath the diaphragm; major left & right lobes + two smaller inferior lobes
Covering	Thin c.t. capsule (Glisson's capsule) + mesothelium of visceral peritoneum; capsule thickens at the hilum , where the portal vein & hepatic artery enter and the right/left hepatic ducts + lymphatics leave



Glisson's capsule overlying hepatocyte plates.

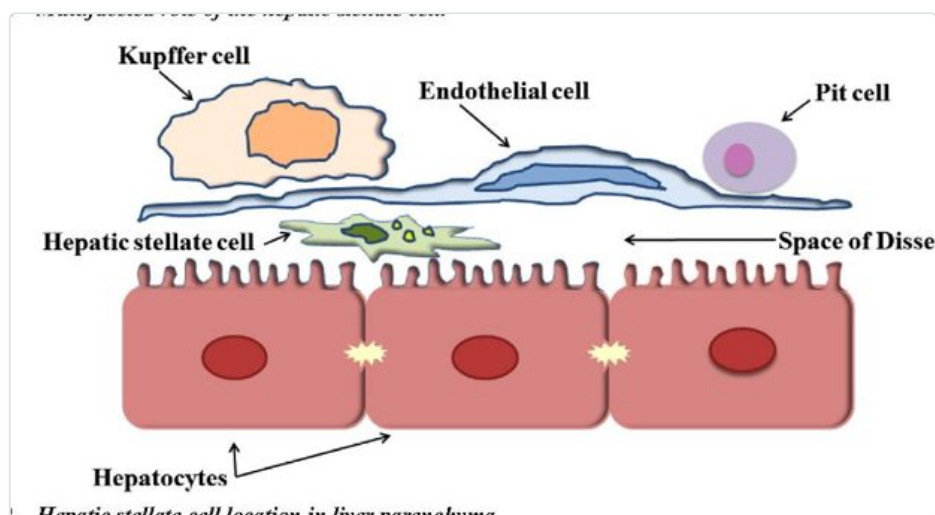
2 Parenchyma & the portal triad

Lobule	Hexagonal-shaped; hepatocytes radiate outward from a central vein
Portal triad	At each of the 6 corners; 3–6 triads per lobule. Consists of: 1) a duct (bile duct system) 2) an arteriole (branch of hepatic artery) 3) a venule (branch of portal vein) 4) lymphatic vessels
Portal canal	The portal triad + surrounding c.t., together



Left: lobules with portal triads at the corners. Right: portal triad close-up — duct, arteriole, venule.

3 Cells in the liver — ID table



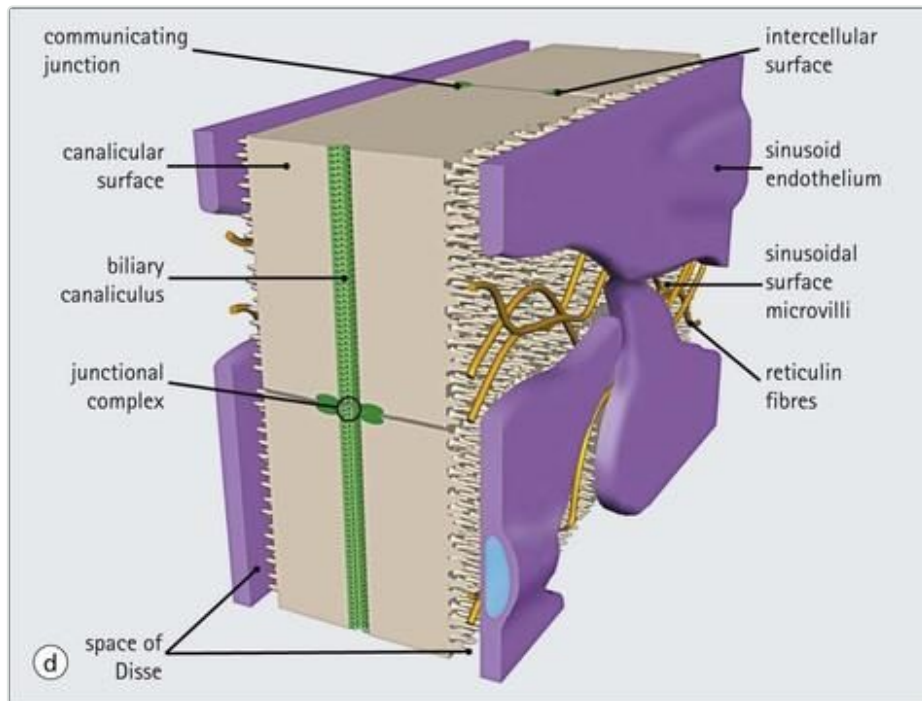
Hepatocytes, sinusoid endothelium, Kupffer cells and hepatic stellate (Ito) cells relative to the space of Disse.

Cell type	Location	Key ID feature	Function
Hepatocyte	Arranged in anastomosing plates; a sinusoid on 2 sides of each cell	Large, polyhedral, 6+ surfaces; eosinophilic cytoplasm rich in mitochondria/organelles	Primary structural & functional liver cell
Sinusoid	Between hepatocyte plates	Irregular, dilated; discontinuous fenestrated endothelium + basement membrane	Blood flows here — separated from hepatocyte surface by the space of Disse (hepatocyte microvilli project into it)
Kupffer cell	Luminal surface of sinusoid endothelium	Monocyte-derived; ovoid nucleus; often contains engulfed material	Phagocytose aged RBCs (recycle heme/iron via ferritin); antigen-presenting; clear bacteria/debris from portal blood; produce ~5% of plasma proteins
Ito (hepatic stellate) cell	Space of Disse	Lipid droplets in cytoplasm	Store vitamin A; major source of collagen/ECM during liver injury; regulate sinusoidal blood flow (contractile); regulate Kupffer cell activity via cytokines



Sinusoids winding between hepatocyte plates.

4 Hepatocyte surface — three specialisations



Sinusoidal, canalicular and intercellular surfaces of the hepatocyte.

1. Sinusoidal surface

Faces the blood sinusoid; microvilli project into the space of Disse — exchanges materials with blood

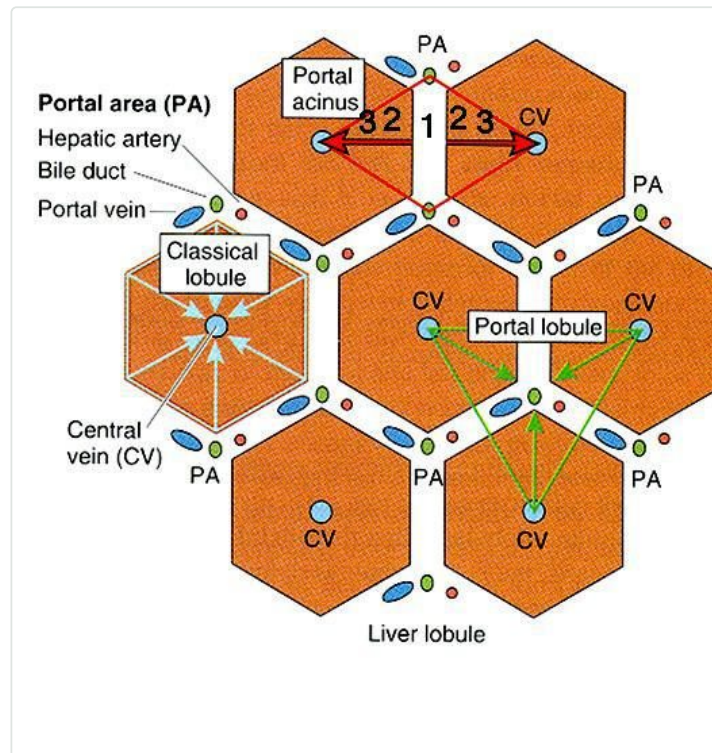
2. Canalicular surface

Longitudinal depression between neighbouring hepatocytes forms the wall of a **bile canaliculus**; irregular microvilli project in — primary site of bile secretion

3. Intercellular surface

Contact area between adjacent hepatocytes; carries the cell junctions that allow exchange between cells

5 Three ways to subdivide the liver



Classic lobule (blood flow), portal lobule (bile flow), and hepatic acinus (metabolic zonation) — three overlapping models of the same tissue.

Model

Shape / landmarks

Based on

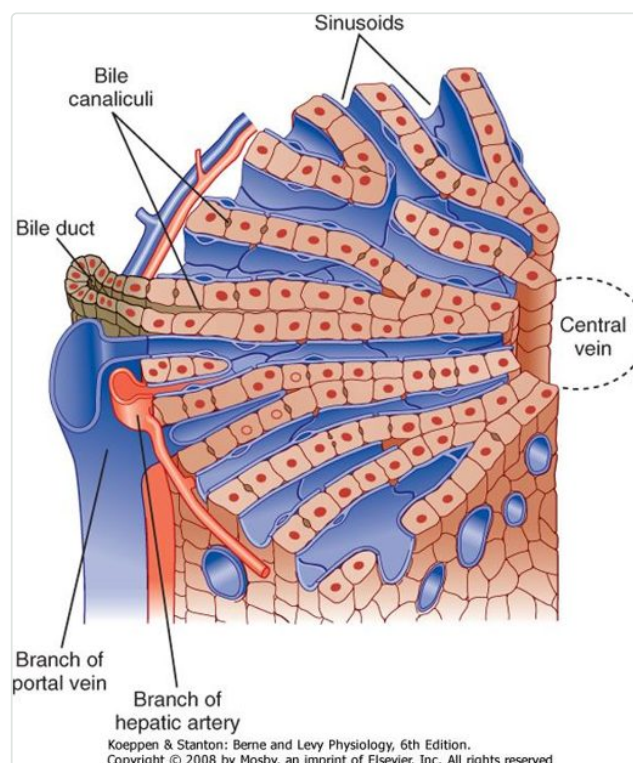
1. Classic lobule	Hexagonal; portal triad at each corner, central vein in the centre	Direction of blood flow
2. Portal lobule	Interlocking triangles; portal triad at the centre, a central vein at each of the 3 corners	Direction of bile flow (opposite to blood)
3. Hepatic acinus	Diamond; 2 central veins + 2 portal triads at its 4 corners, split into 3 zones by distance from the triad	Oxygen/nutrient/toxin gradient as blood crosses the sinusoids

Acinus zones: **Zone I** (nearest the portal triad) has the highest oxygen, nutrient and toxin concentration; **Zone III** (nearest the central vein) has the lowest — the most ischaemia-prone zone.

6 Blood circulation in the liver

Portal vein — 80%	Conveys blood from pancreas, spleen & intestine; oxygen-poor, nutrient-rich
Hepatic artery — 20%	Higher oxygenated blood

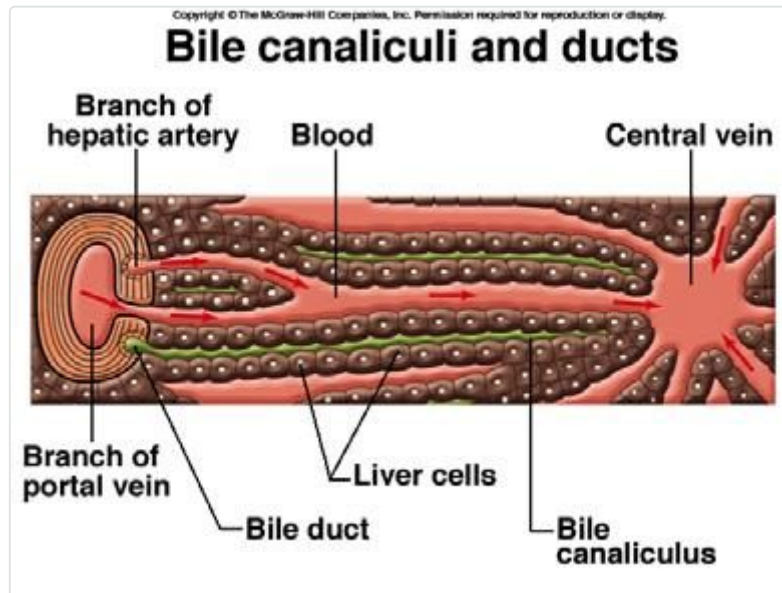
Both empty into the **sinusoid** → **central vein** → right/left **hepatic vein** → **inferior vena cava**.



Bile canaliculi run opposite to sinusoidal blood flow, both converging on the portal triad / central vein axis.

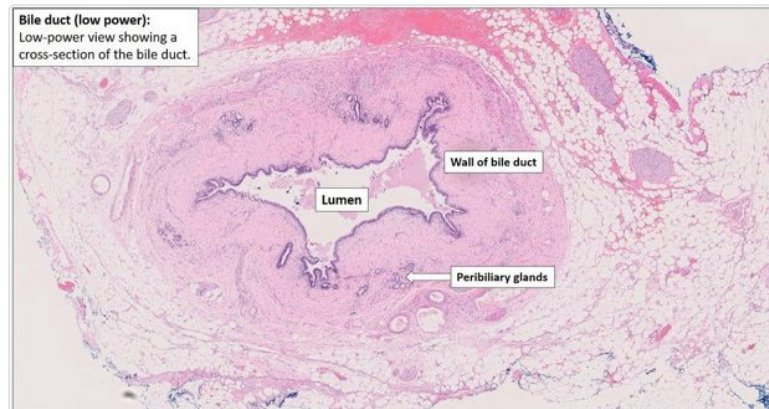
7 Bile duct system

Bile canaliculus (bounded only by the plasma membranes of two adjacent hepatocytes — no distinct wall) → bile ductules (**canals of Hering**, cuboidal) → bile ducts (cuboidal→columnar, in the portal triad) → right & left hepatic ducts leave the liver → + cystic duct (from gallbladder) → common hepatic duct → common bile duct → + main pancreatic duct → **ampulla of Vater** → duodenum, gated by the **sphincter of Oddi**.



Bile canaliculi drain opposite to sinusoidal blood flow, toward the portal triad.

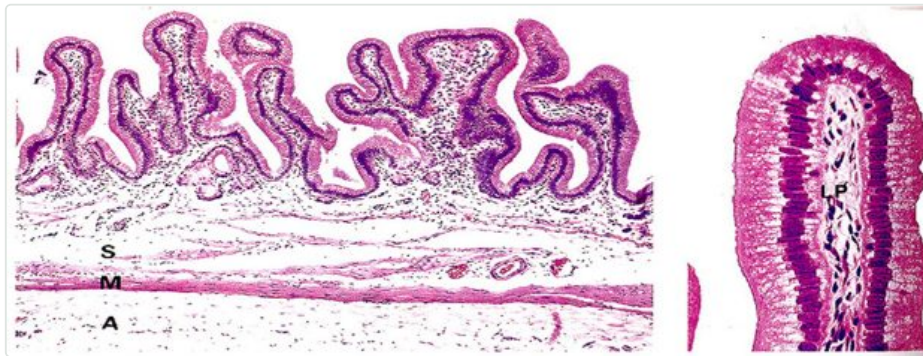
Bile composition	Water, electrolytes, bile acids, phospholipids, cholesterol, bilirubin
Bile acid source	~90% reabsorbed from the intestine; ~10% synthesised in hepatocyte smooth ER
Bile acid function	Emulsify lipids in the digestive tract — easier digestion by lipase & absorption
Duct histology	Hepatic, cystic & common bile ducts: simple columnar epithelium + thin muscularis; muscularis thickens near the duodenum, forming the sphincter of Oddi at the duodenal papilla



Bile duct cross-section — lumen, wall, and peribiliary glands.

8 Extrahepatic biliary apparatus — the gallbladder

Shape/location	Pear-shaped sac under the liver
Function	Stores & concentrates bile; releases it via the bile duct; secretion regulated by cholecystokinin (CCK)
Wall layer	Histology
Mucosa	Highly folded; simple columnar cells with microvilli; no goblet cells ; supported by lamina propria
Submucosa	Similar to rest of GI tract — blood vessels, lymphatics, nerves
Muscularis	2–3 irregular layers of smooth muscle
Serosa / adventitia	Mesothelium, except adventitia where the gallbladder contacts the liver



Gallbladder mucosa — deep folds of simple columnar epithelium with lamina propria (LP).

P Must-remember pearls

- Portal triad = bile duct + hepatic arteriole + portal venule ($\pm$ lymphatics), at the lobule's corners; with surrounding c.t. it's called the **portal canal**.
- **Kupffer cells** = resident macrophages on the sinusoid lining. **Ito (stellate) cells** = space of Disse, store vitamin A, become the collagen source in liver injury.
- Hepatic acinus zones: **Zone I** (periportal) = most oxygen/nutrients; **Zone III** (pericentral) = least — most vulnerable to ischaemia.
- 80% of liver blood = portal vein (nutrient-rich, low O_2); 20% = hepatic artery (high O_2) — both drain into the sinusoids.
- Bile path: canaliculus $\rightarrow$ canal of Hering $\rightarrow$ bile duct $\rightarrow$ hepatic duct $\rightarrow$ (+ cystic duct) $\rightarrow$ common bile duct $\rightarrow$ (+ pancreatic duct) $\rightarrow$ ampulla of Vater $\rightarrow$ duodenum.

Q Test yourself

1. Which structure is found at each of the six corners of a classic hepatic lobule?

- | | |
|-------------------------|------------------------|
| A) Central vein | B) Portal triad |
| C) Kupffer cell cluster | D) Islet of Langerhans |

2. A liver cell in the space of Disse, storing vitamin A in lipid droplets and becoming a major collagen source during injury, is the:

- | | |
|------------------------|----------------------|
| A) Kupffer cell | B) Hepatocyte |
| C) Ito (stellate) cell | D) Centroacinar cell |

3. In the hepatic acinus model, blood in zone I sinusoids compared with zone III is:

- | | |
|-----------------------------|-----------------------------------|
| A) Lower in oxygen | B) Higher in oxygen and nutrients |
| C) Higher in bilirubin only | D) Identical |

4. The wall of the bile canaliculus is formed by:

- | | |
|---------------------------------|---|
| A) A distinct epithelial lining | B) The plasma membranes of two adjacent hepatocytes |
| C) Cuboidal duct cells | D) Fenestrated endothelium |

5. Bile release from the gallbladder is triggered mainly by:

- | | |
|--------------------------|-----------------|
| A) Secretin | B) Gastrin |
| C) Cholecystokinin (CCK) | D) Somatostatin |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1. B — Portal triad** — duct + arteriole + venule ($\pm$ lymphatics) at each of the 6 lobule corners.
- 2. C — Ito (stellate) cell** — space of Disse; vitamin A storage; collagen source in fibrosis.
- 3. B — Higher oxygen and nutrients** — zone I is nearest the portal triad, before the sinusoidal gradient drops toward zone III.
- 4. B — Plasma membranes of two hepatocytes** — no distinct duct wall until the canal of Hering.
- 5. C — CCK** — released from duodenal mucosa in response to fat/protein in chyme.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam